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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
Faculty Name	Dr. Sukanya
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Roll Number	2520030128
Branch	CSE
Course Outcome	1

Case Study Topic: Stack Overflow Reputation Index – BST Range Query using AVL Tree

Word Done Summary:

In this case study, we analyzed the use of an **AVL Tree (Balanced Binary Search Tree)** for performing efficient range queries on Stack Overflow user reputation scores.

Stack Overflow maintains millions of users, each associated with a reputation score. The leaderboard system frequently needs to answer queries such as:

"List all users whose reputation lies between 10,000 and 50,000."

The users are stored in an AVL Tree keyed by reputation score. The objective is to evaluate whether AVL Tree range scanning is efficient and compare it with an alternative approach using a sorted array with binary search.

Implementation Details:

1. AVL Tree construction.
2. Range query traversal.
3. In-order scanning within bounds.
4. Complexity analysis.
5. Comparison with sorted arrays.
6. Suitability for static and dynamic workloads.

Result:

Input Reputation Scores

5000

12000

15000

25000

30000

50000

60000

70000

Query:

[10000,50000]

Output:

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Users in Reputation Range:
12000 15000 25000 30000 50000
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GitHub Link:

https://github.com/saisharan134/DSA-2-Case-Studies

Student Signature	Faculty Signature
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